An Empirical Study of CDGS-Based Molecular Generation on QM9: Effects of Sampling Steps and Lightweight Chemistry-Aware Post-Processing

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Abstract

This study investigates how the diffusion sampling budget affects molecular graph generation in discrete graph diffusion models. Using the VPSDE-based CDGS framework as a fixed baseline, we conduct a controlled empirical evaluation on the QM9 dataset with a pretrained checkpoint (`checkpoint_200`).

Our experiments focus on the effect of the sampling step budget $T \in \{25, 50, 100, 200\}$ on generation quality, diversity, and computational efficiency. In addition, we examine a lightweight chemistry-aware post-processing strategy, denoted as *filterplus*, which filters generated molecules using RDKit sanitization, atom-type constraints (C, N, O, F), connectivity checks, and simple valence rules.

Based on 5,000 generated molecules per setting, we evaluate the results using metrics including Validity, FCD/Test, Novelty, Unique ratio, Retention ratio, and failure breakdown. The results show that reducing T substantially improves sampling efficiency, but excessive compression leads to markedly worse distributional quality and a much higher rate of chemically problematic outputs. By contrast, larger step budgets consistently produce better-quality and less repetitive samples, although at higher computational cost. The proposed lightweight filtering strategy provides modest improvements in the retained set, but its effect mainly comes from discarding invalid molecules rather than correcting them.

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1 Introduction

[Primary author: Yuefeiyang Li]

This project studies molecular graph generation, at the intersection of graph generative modeling, diffusion models, and AI for science. The goal of this line of research is to learn the distribution of molecular data and generate new molecules that are structurally reasonable and potentially useful in practice. Since molecules can naturally be represented as graphs, graph neural networks and graph generative models have become important tools for molecular modeling Gilmer et al. [2017]. Against this backdrop, molecular generation is not only a theoretical generative task, but is also closely related to practical applications such as drug discovery, materials design, and candidate molecule search. The original CDGS paper similarly treats learning molecular graph distributions and generating high-quality new molecules as a fundamental problem in drug discovery and materials science, and emphasizes that high generation quality and fast sampling are both important goals, even though they cannot always be achieved at the same time Huang et al. [2023].

Compared with sequence-based molecular representations, graph-based representations preserve structural information more directly, and many recent methods therefore model molecules as graphs. In our project, we consider 2D molecular graph generation, where each molecule is represented as a discrete graph with nodes corresponding to atom types and edges corresponding to bond types. This representation is consistent with the basic modeling idea of CDGS, which provides a natural interface to incorporate simple chemical rule checks in later stages.

In recent years, diffusion models have demonstrated increasing potential in the generation of molecular imagesHo et al. [2020], Song et al. [2020], Jo et al. [2022]. Unlike autoregressive methods that build graphs piece by piece, diffusion models recover the entire structure from noise through a progressive 'noising-denoising' process, offering superior training stability and modeling flexibility. Although existing diffusion-based molecular graph methods have demonstrated strong generation ability, they still face two major challenges Jo et al. [2022], Huang et al. [2023]. First, there remains a gap between the quality of the generated samples and the true data distribution. Second, sampling is often slow and inference can be computationally expensive, especially when the generation process relies on a large number of steps or additional correction procedures.

Against this backdrop, our project does not aim to propose an entirely new molecular generation model. Instead, we focuses on two more specific and practically motivated questions: whether a discrete diffusion model that is already capable of generating molecules can still maintain sufficiently good generation quality when the number of sampling steps is reduced and whether lightweight chemistry-aware post-processing can further make the generated results more valid and reliable. More specifically, our project is centered on two core research questions. First, how does the sampling budget, that is, the number of diffusion steps, affect generation quality and efficiency under the none setting, as measured by FCD/Test, Runtime, Novelty, and diversity-related statistics? Second, can lightweight chemistry-aware post-processing, denoted as filterplus, further improve the reliability of the generated molecules without causing an excessive loss of retained diversity? These two questions involve a trade-off among sampling efficiency, chemical validity, and diversity. Reducing the number of sampling steps may substantially improve runtime, but it can also lead to non-linear degradation in sample quality. Similarly, post-processing filters may improve final validity, but they often do so by discarding invalid samples, making it difficult to determine whether the gain in reliability justifies the loss in diversity.

These two questions are important because they address two practical requirements in molecular generation at the same time: efficiency and reliability. From the perspective of efficiency, diffusion models usually require a relatively large number of denoising steps Ho et al. [2020], Song et al. [2020]. Although using more steps often makes the generation process more stable, it also leads to a higher sampling cost. In practical molecular design tasks, it is not enough

for a model merely to generate molecules; it also needs to do so efficiently. From a reliability perspective, the generated graphs do not necessarily satisfy basic chemical rules by default Huang et al. [2023]. For example, some atoms may have unreasonable valence patterns, or the resulting structures may fail RDKit sanitization. In other words, even if a model performs well in a statistical sense, it may still produce a large number of chemically invalid molecules. For this reason, it is worth testing whether simple post-processing can improve the final outputs. In real-world molecular design, an ideal generative model is more than just a source of chemically sound candidates; it must also be computationally practical, maintaining high sampling throughput under a limited computational budget.

In terms of method selection, our project is based primarily on the CDGS baseline Huang et al. [2023]. We chose CDGS because it is highly aligned with the research questions of our project. First, CDGS itself is a discrete graph diffusion framework specifically designed for molecular graph generation rather than serving as a general-purpose graph diffusion baseline. Second, the original CDGS paper explicitly emphasizes that the method can still maintain good generation quality under a limited number of sampling steps, and it specifically discusses its few-step and fast-sampling capability Huang et al. [2023]. This makes CDGS particularly suitable for a step-budget sweep: once the trained model is fixed, we can directly compare the change in speed and quality by systematically varying the number of sampling steps.

Based on this setup, on QM9, we use CDGS as a fixed baseline to study how sampling budget and lightweight chemistry-aware post-processing affect molecular graph generation, and to analyze the trade-offs among distributional quality, diversity, reliability, and runtime. More specifically, we focus on four step settings $T \in \{25, 50, 100, 200\}$, and compare two settings, *none* and *filterplus*, where the latter applies lightweight post-processing checks such as RDKit sanitization, atom-type filtering, connectivity screening, valence checks, and size constraints. Under this design, the whole project can be viewed as a small but clear two-dimensional experimental matrix, which helps to keep the research question focused and also makes the later result analysis more straightforward.

Our project also has several clear scope limitations. First, experiments are restricted to QM9 rather than extended to larger and more complex molecular datasets, so the resulting conclusions mainly apply to the setting of small-molecule graph generation. Second, our project uses a pretrained CDGS model without redesigning the training objective or introducing complex chemical repair algorithms. As a result, the study is better understood as a systematic experimental analysis of an existing generative framework rather than as a proposal of a new model architecture. Third, Novelty is still retained as a common evaluation metric, yet the original CDGS paper points out that the novelty metric on QM9 is itself somewhat debatable Huang et al. [2023]. Since QM9 is close to a relatively complete collection of small molecules under specific constraints, some molecules that appear novel do not necessarily indicate better distribution learning. We therefore interpret our findings as results specific to this setup rather than general conclusions for all molecular generative models. They are not intended as universal laws applicable to every molecular generative architecture or every chemical space, but rather as a focused look at the trade-offs within this particular regime.We therefore interpret our findings as results specific to this setup rather than general conclusions for all molecular generative models.

Overall, this project asks two practical questions: how much quality is lost when sampling is made faster, and whether very lightweight post-processing can improve chemical validity at acceptable cost. To answer these two questions, we systematically compare generation results under different sampling step settings and constraint strategies on a fixed CDGS baseline, and ultimately examine the trade-offs among speed, distributional quality, reliability, and diversity.

2 Background

[Primary author: Shiya Feng]

2.1 Dataset Overview

The primary dataset used in this study is the QM9 dataset, a standard benchmark dataset in computational chemistry and machine learning. This dataset contains approximately 134,000 stable small organic molecules, each containing up to nine heavy atoms (carbon, oxygen, nitrogen, and fluorine). These data were not obtained through physical measurements, but rather by extracting a subset from the GDB-17 chemical system and labeling its properties using high-fidelity quantum mechanical simulations. The QM9 dataset is well-suited for answering our research questions because it provides a validated, standardized baseline of three-dimensional molecular structures and properties, which facilitates the training and evaluation of our chosen network architecture. A more comprehensive breakdown of the dataset's features and processing steps is detailed in Chapter 4.

2.2 Discrete Graph Diffusion

Generative Models for Graphs Previous tasks of de novo molecular generation primarily relied on representing molecules as one-dimensional SMILES strings or utilized autoregressive and variational frameworks for processing. Early graph generation models (e.g., GraphRNN You et al. [2018]) modeled the generation process as a sequential procedure that involves the addition of elements node-by-node. Other approaches, such as employing frameworks such as Graphite Grover et al. [2019], leveraged Variational Autoencoders (VAEs) and Message Passing Neural Networks to iteratively reconstruct graph structures. Although these models achieved a certain degree of success, they often struggled to effectively capture complex and long-range topological dependencies without triggering severe training instability or becoming ensnared in slow sequential sampling bottlenecks.

The introduction of Denoising Diffusion Probabilistic Models (DDPMs) Ho et al. [2020] and their unification within the framework of stochastic differential equations Song et al. [2020] have revolutionized the field of generative modeling. These models achieve state-of-the-art performance by utilizing continuous Gaussian noise to progressively corrupt data and training neural networks to reverse this process. However, applying continuous Gaussian noise to categorical data, such as discrete atom types (e.g., carbon, nitrogen) and bond types (e.g., single bonds, double bonds), can disrupt the inherent sparsity of molecular graphs and leads to the generation of physically meaningless intermediate states.

Shift to Discrete Diffusion Spaces To address the mismatch between continuous noise and categorical data, Austin et al. [2021] introduced discrete denoising diffusion probabilistic models (D3PMs). Unlike methods that add Gaussian noise, D3PMs uses categorical transition matrices to iteratively flip data states, such as pixel colors or text tokens based on a predefined Markov chain.

This theoretical breakthrough paved the way for research into diffusion models tailored specifically to graph structures. Haefeli et al. [2022] demonstrated that diffusion models designed for graph structures inherently benefit from a discrete state space, as such a space preserves the sparsity of the graph structure throughout the forward noise-addition process.

Building upon this foundation, Vignac et al. [2022] proposed DiGress: a discrete diffusion denoising model for graph generation. DiGress defines a forward Markov process comprising a sequence of discrete graph editing operations. Specifically, this process involves the independent addition or deletion of edges, as well as the mutation of node categories. Subsequently, the system trains a Graph Transformer network to directly predict the clean categorical graph structure

from noisy inputs. The primary strengths of DiGress lie in its mathematically rigorous discrete diffusion framework and its robust graph generation capabilities.

Alternative Approaches DiGress follows the design principles of the General Graph. It treats the molecule purely as a topological entity, defined by the adjacency matrix A and the characteristic matrix X, while digressing and striving to maximize the likelihood function with permutation equivariance. This approach represents a distinct departure from competing methods, such as Graph Score-based diffusion generative modeling through SDEs (GDSS), proposed by Jo et al. [2022]. GDSS attempts to model the joint distribution of nodes and edges using a continuous-time stochastic differential equations system (SDEs), relying on "score-matching" techniques to handle the intricate dependencies between nodes and edges. Although continuous methods like GDSS are powerful, the discrete transition matrices employed in DiGress often demonstrate superior performance regarding the local control of graph structures.

Key Limitations Although DiGress has mathematically elegant properties, its generic nature presents significant obstacles when applied specifically to the field of drug discovery. A primary limitation is the model's propensity to generate invalid molecules. Lacking explicit chemical inductive biases such as strict valence constraints or thermodynamic rules, DiGress frequently predicts structures that are chemically impossible, such as carbon atoms forming five bonds.

Furthermore, the process of modeling edges in DiGress can be computationally expensive. The model requires calculating transition probabilities for every potential edge within a dense $N \times N$ adjacency matrix. For large molecular graphs, this creates an extremely vast search space, one predominantly occupied by "empty bonds", resulting in prohibitively high computational overhead. Recent research efforts (e.g., autoregressive diffusion models for graph generation by Kong et al. [2023]) have attempted to mitigate this issue by dynamically masking and unmasking nodes; however, computational cost remains a persistent challenge in the context of parallel graph generation tasks.

Ultimately, while diffusion models are capable of achieving high accuracy, they often operate at a relatively slow pace; yet what molecular design tasks truly demand are candidate molecules that are both rapidly generated and chemically valid.

2.3 Conditional Diffusion on Discrete Graph Structures (CDGS)

When applied to molecular data, standard continuous diffusion models encounter significant theoretical hurdles. Embedding discrete graphs into a continuous space, and injecting Gaussian noise into node features and adjacency matrices which disrupts the inherent topological sparsity of molecular graphs Chen et al. [2024]. This continuous-space corruption process yields dense and noise-laden structures, rendering key chemical properties, such as bond rigidity, valence rules, and the number of ring systems mathematically undefined during intermediate generation steps.

Furthermore, graph models based purely on real-valued representations struggle during the denoising phase to accurately extract local structural motifs from noise-corrupted matrices; this results in generated molecular distributions that deviate significantly from those of real-world molecular datasets. To overcome these challenges, researchers have developed discrete diffusion frameworks that can operate directly within categorical spaces, thereby preserving the discrete nature of atoms and chemical bonds. Against this backdrop, we have selected the conditional diffusion model for discrete graph structures (CDGS) as our primary benchmark model. This model is specifically engineered to effectively handle the unique discrete topological constraints to the chemical domain while simultaneously maintaining highly competitive performance levels on molecular benchmark tasks such as QM9 Huang et al. [2023].

Molecule-Tailored Architecture CDGS differentiates itself from generic graph generators through an architecture highly tailored to molecular properties that ingeniously integrates

continuous stochastic differential equations with a discrete graph conditioning mechanism. The model employs stochastic differential equations to construct a forward graph diffusion process, applying this process simultaneously to both the structural topology of the graph and its intrinsic node and edge features. To effectively handle the discrete nature of graphs without relying on specialized discrete state spaces, CDGS introduces a simplified quantization mechanism Vignac et al. [2022]. Complementing the standard matrices for node and edge features, this mechanism incorporates a set of discrete one-bit variables that explicitly indicate the presence or absence of chemical bonds. During the continuous forward SDE evolution, these discrete "edge existence" variables are converted into real-valued representations by applying strict quantization thresholds. Conversely, during the reverse generation phase, these discrete graph structures are decoded first and serve as mandatory conditions that must be adhered to at every subsequent sampling step. This conditioning mechanism ensures that the entire generative process remains consistently anchored to valid graph topologies.

Moreover, CDGS employs a specially designed hybrid graph noise prediction model. Unlike previous architectures that used independent and unrelated networks to denoise nodes and edges separately, CDGS's noise prediction model requires an interaction between the representations of nodes and edges, which are derived from both continuous real-valued matrices and decoded discrete structures. This unified interaction mechanism enables the model to simultaneously extract high-level global contextual information and fine-grained local node-edge dependencies from noisy intermediate graph states. In this way, CDGS successfully overcomes the strict chemical constraints that standard graph neural networks (GNNs) often struggle to meet, such as maintaining local chemical valence constraints, maintaining realistic proportions of atom types, and reproducing global structural patterns at the molecular level.

Alternative Baselines To fully understand why we selected CDGS as the primary benchmark, it must be examined within a comparative context with other contemporary discrete diffusion models. The aforementioned DiGress is an outstanding general-purpose discrete benchmark model. DiGress progressively disrupts and denoises the graph structure through discrete editing operations such as adding or removing edges and modifying discrete node properties, and its operation relies entirely on a permutation-equivariant graph Transformer network. Although DiGress excels in handling arbitrary graph topologies, it does not explicitly customize or optimize for stringent heuristic constraints in chemistry, unlike CDGS. Other recently emerging architectures introduce unique mechanisms to address the sparsity of graph structures. EDGE model fully exploits the sparsity of graph structures by defining a generation mechanism that reverses the "edge removal" process Chen et al. [2023]. This model prioritizes the preservation of the degree of the node and improving computational efficiency over a comprehensive consideration of chemical validity. Meanwhile, DISCO model, by introducing a continuous time Markov chain, constructs a graph diffusion process using discrete states and continuous time, thus achieving a flexible trade-off in sampling strategies Xu et al. [2024].

In the related literature, some more complex models with in-depth optimization for specific domains have also emerged. MG-DIFF model employs a discrete diffusion masking and substitution strategy that combines graph-filling techniques, enabling dynamic control of molecule size during generation Zhang et al. [2025]. MolHIT model integrates a hierarchical discrete diffusion framework, aiming to directly inject high-level chemical prior knowledge and decoupled atomic encoding into the diffusion process. Although these cutting-edge models continue to push the boundaries of technology in the field of conditional generation, for this project, CDGS remains the most ideal and fundamental benchmark model because it embodies the core principles of a discrete-continuous hybrid diffusion mechanism tailored to molecular structures. In numerous datasets containing synthetically feasible small organic molecules, CDGS demonstrates highly competitive performance without the heavy architectural overhead of hierarchical or projection operator-based models; for this reason, it has become the ideal platform for the systematic

ablation studies planned in our methodology.

As mentioned earlier, our experiments employed a fixed pre-training checkpoint. Here, we explain the learning algorithm and loss function used to optimize this CDGS model. The model is trained using a standard gradient-based optimization algorithm to minimize the combined cross-entropy loss. Since the discrete diffusion framework trains the network to predict the original clean graph x_0 from any noisy intermediate state x_t, the primary training objective is to maximize the log-likelihood of the true classification data. Specifically, the loss function is decomposed into two cross-entropy terms, corresponding to node type (atoms) and edge type (bonds), respectively:

$$\mathcal{L}(\theta) = \mathbb{E}_{t\sim\mathcal{U}(1,T),x_0,x_t}\left[\mathrm{CE}(X_0, \hat{X}_\theta(x_t,t)) + \lambda \mathrm{CE}(E_0, \hat{E}_\theta(x_t,t))\right]$$

where CE represents the cross-entropy loss, X_0 and E_0 represent the true discrete node matrix and edge matrix, respectively, $\hat{X}_\theta$ and $\hat{E}_\theta$ are the predicted probability distributions of the Graph Transformer for the clean graph, and λ is a balancing hyperparameter. This learning algorithm ensures that the network can recover the sparse, classifiable topology required for effective molecular structures.

Limitations Despite its robust, molecular-oriented design, CDGS suffers from a critical limitation, which is the starting point of our project's core investigation: it is highly sensitive to the selection of samplers, noise schemes, and step budgets. During the sampling phase, CDGS efficiently performs graph sampling using an ordinary differential equation solver, particularly the Euler-Maruyama method, bypassing the computationally intensive Langevin correction step found in previous score-based models. The authors of CDGS demonstrate that approximately 200 network evaluation steps are sufficient to generate high-fidelity samples .

However, this reliance on a strictly predefined sampling budget creates a significant operational bottleneck. The exact number of diffusion steps drastically determines the quality of the final generated molecules, establishing a strict trade-off between inference speed and generation quality. Reducing the sampler budget to accelerate generation often results in the model being affected by accumulated denoising errors Boget [2025]. In the discrete and mixed diffusion processes, intermediate noise states heavily depend on previous steps. As the number of steps decreases, the network fails to accurately map the probability flow ordinary differential equations, leading to excessive error accumulation during reverse denoising.

Therefore, the step limit results in incomplete structural recovery, generating molecular graphs that violate fundamental chemical rules, and showing impossible valence states Cardei et al. [2025]. Conversely, simply increasing the number of steps to ensure validity can severely impact the runtime per sample, thus diminishing the model's practicality in high-throughput molecular design, where rapid generation is crucial. This sensitivity suggests that, while CDGS is mathematically rigorous, it is not inherently robust to hyperparameter variations during inference. Therefore, understanding how variations in the number of steps $T \in \{25, 50, 100, 200\}$ directly affect the Pareto front for validity, uniqueness, novelty, and processing speed remains an unsolved problem.

3 Methods

[Primary author: Shiya Feng]

This chapter describes the mathematical framework, computational architecture, and experimental design for generating two-dimensional molecular graphs via discrete diffusion. We primarily build upon the discrete graph structure continuous-time diffusion (CDGS) baseline within the variance-preserving stochastic differential equations (VPSDE) framework Huang et al. [2023], and integrate theoretical insights from the discrete denoising diffusion probabilistic model (D3PM) Austin et al. [2021] to formalize the molecular representation. Furthermore, we outline the temporally nonhomogeneous Markov process driving diffusion, describe the topological mechanism of the graph converter architecture, and define the mathematical formulations for experimental constraints and evaluation metrics. Importantly, to rigorously isolate the effects of the diffusion sampling step, we use a frozen, pre-trained generator instead of retraining the model from scratch.

3.1 Feature Encoding

To leverage probabilistic generative models for 0-to-1 chemical discoveries, molecules must be finely mapped to a structured mathematical format compatible with graph-based neural networks. Gilmer et al. [2017] Historically, generative models have used one-dimensional string representations, such as the SMILES model. However, as discussed in Chapter 2, string-based representations are often inefficient due to their syntactic fragility, and changes to a single character can render an entire sequence chemically meaningless. To overcome this limitation and directly model the topological reality of chemical structures, we conceptualize molecules as two-dimensional, undirected graphs with node and edge attributes.

Suppose a specific molecule in a dataset is defined as a mathematical graph $G = (V, E)$, where V represents the set of nodes (atoms) with cardinality $|V|$. N represents the set of edges (chemical bonds), and E represents the set of edges. For computational batching, N is typically padded to the maximum number of atoms present in the target dataset. In our experiments using the QM9 dataset Ramakrishnan et al. [2014], we imposed a strict constraint that the maximum number of atoms per molecule could not exceed 9. Table 1 below compares the dimensions and categories of nodes and edges. The topological structure of the graph and its corresponding chemical features are fully described by two distinct feature tensors:

Node Feature Matrix (X) Atoms are represented by the matrix $X \in \{0, 1\}^{N \times d_n}$, where d_n represents the dimension of the atom type feature space. In our experiments, we strictly restrict the generated atom types to a subset of organic elements and set the state space to $d_n = 4$. These dimensions correspond precisely to the discrete chemical elements: carbon (C), nitrogen (N), oxygen (O), and fluorine (F). Hydrogen atoms are implicitly handled during generation to reduce topological complexity and computational overhead. Each row vector $x_i \in X$ corresponding to node v_i is a strictly one-hot encoded vector satisfying $\sum_{k=1}^{d_n} x_{i,k} = 1$.

Edge Feature Tensor (E) The chemical bonds connecting atoms are encoded in a three-dimensional adjacency tensor $E \in \{0, 1\}^{N \times N \times d_e}$, where $d_e = 4$. This tensor captures a specific bond order between any two nodes v_i and v_j. Discrete categories are defined as single bonds, double bonds, triple bonds, and structurally "bonded" categories. Explicitly including the "bonded" category allows the neural network to learn and generate sparse topologies of graphs during diffusion without manually designing structural masks or heuristic pruning Vignac et al. [2022]. For any biologically or chemically valid undirected graph, this tensor maintains symmetry such that $E_{i,j,k} = E_{j,i,k}$ for all $k \in \{1, 2, 3, 4\}$.

Table 1: Summary of Node and Edge Feature Encodings

Feature Type	Discrete Categories	Dimension
Node Matrix (X)	Carbon (C), Nitrogen (N), Oxygen (O), Fluorine (F)	$d_n = 4$
Edge Tensor (E)	No bond (0), Single (1), Double (2), Triple (3)	$d_e = 4$

3.2 Discrete Diffusion Framework

Standard Gaussian diffusion models have achieved state-of-the-art results in continuous domains such as image and audio generation, relying on mapping continuous data to isotropic Gaussian noise Ho et al. [2020]. However, molecular graphs are inherently discrete topological structures. Smoothing interpolations between categorized carbon and oxygen atoms in continuous space lack physical meaning and often generate invalid intermediates. To address this, we employ a discrete diffusion framework modeled as a time-nonhomogeneous Markov chain with a categorized transition matrix Austin et al. [2021] Jo et al. [2022]. Our chosen baseline model, CDGS, explicitly utilizes a variance-preserving stochastic differential equation scheduling algorithm to map the discrete structure to continuous time while preserving the signal variance.

3.2.1 Pre-trained Checkpoint Evaluation

To ensure all comparative evaluations are performed with mathematically identical generative capabilities, instead of training the graph generation model from scratch, we use a fixed pre-trained model checkpoint (labeled **checkpoint_200**). By freezing the neural network weights θ, any differences observed in the generated molecular distribution or overall effectiveness are entirely and necessarily attributable to adjustments in the step size budget T.

3.2.2 Forward Noise Process (q)

The forward noise process gradually destroys the structure of a clean, chemically valid graph $G_0 = (X_0, E_0)$ into purely discrete noise states over fixed T time steps. Let x_t represent a categorical random variable at time $t \in \{1, \ldots, T\}$, representing a specific node or edge type. The random noise from step $t-1$ to t is controlled by the transition matrix Q_t, where the specific probability of transitioning from categorical state i to state j is given by the matrix element $[Q_t]_{ij}$.

The forward transition probability distribution is formulated as a categorical distribution conditioned on the preceding state:

$$q(x_t|x_{t-1}) = \{Cat\}(x_t; p = x_{t-1}Q_t)$$

Since this process strictly satisfies the Markov property, the cumulative product of the transition matrices can be used to efficiently compute the distribution of any noisy state x_t under the condition of the original clean data x_0. Let $\bar{Q}_t = Q_1 Q_2 \ldots Q_t$ denote this cumulative transition matrix. Then the marginal distribution can be derived as:

$$q(x_t|x_0) = \{Cat\}(x_t; p = x_0\bar{Q}_t)$$

3.2.3 Reverse Denoising Process (p_θ)

The primary objective of the generative model is to approximate the true reverse trajectory, $q(x_{t-1}|x_t, x_0)$. Because the true original graph x_0 is unknown during inference, we parameterize the reverse process using the frozen deep neural network $p_\theta(x_{t-1}|x_t)$.

Rather than training the network to predict the microscopic transition from x_t to x_{t-1} directly, the network predicts the completely clean graph x_0 at every timestep. Let the neural

network output a predicted probability distribution over the original clean states, denoted as $p_\theta(\hat{x}_0|x_t)$. Using Bayes' theorem and the known probabilities of the forward transitions, the inverse transitions can be computed by marginalizing all possible original states $\hat{x}_0$:

$$p_\theta(x_{t-1}|x_t) = \sum_{\hat{x}_0} q(x_{t-1}|x_t, \hat{x}_0) p_\theta(\hat{x}_0|x_t)$$

where the true posterior transition $q(x_{t-1}|x_t, \hat{x}_0)$ is handled parsingly and defined as:

$$q(x_{t-1}|x_t, \hat{x}_0) = \frac{q(x_t|x_{t-1})q(x_{t-1}|\hat{x}_0)}{q(x_t|\hat{x}_0)} = \frac{(x_{t-1}Q_t x_t^\top)(\hat{x}_0 \bar{Q}_{t-1} x_{t-1}^\top)}{\hat{x}_0 \bar{Q}_t x_t^\top}$$

3.3 Graph Transformer

To effectively approximate the reverse denoising process $p_\theta(\hat{x}_0|x_t)$, the underlying architecture relies on a highly parameterized Graph Transformer Ying et al. [2021]. This structure combines the inductive biases of local message passing algorithms with the global receptive field of multi self-attention mechanisms to process the specific discrete graph features X and E.

3.3.1 Message Passing

Our graph transformer improves upon the standard attention mechanism by injecting edge features into the computation of attention weights. Let $h_i^{(l)}$ denote the hidden feature vector of node i in layer l of the network. The network computes attention by projecting the hidden state onto the query matrix (Q), key matrix (K), and value matrix (V) using the learned weight matrices W_Q, W_K, W_V:

$$q_i = h_i^{(l)} W_Q, \quad k_j = h_j^{(l)} W_K, \quad v_j = h_j^{(l)} W_V$$

To incorporate the topology, the class edge features e_{ij} connecting nodes i and j are embedded into a continuous vector space and added as basic bias terms to the computation of the attention score. The edge conditional attention weight α_{ij} between node i and node j is calculated as follows:

$$\alpha_{ij} = \text{softmax}_j \left(\frac{q_i \cdot k_j^\top + \text{MLP}(e_{ij})}{\sqrt{d_k}} \right)$$

where d_k is the dimension of the key vector. The updated node representation is a weighted sum of the value vectors of all nodes in the graph, fused using a multilayer perceptron (MLP), and finally stabilized by layer normalization:

$$h_i^{(l+1)} = \text{LayerNorm} \left(h_i^{(l)} + \text{MLP} \left(\sum_{j=1}^{N} \alpha_{ij} v_j \right) \right)$$

3.4 Experimental Design Method

The core scientific inquiry of this project is to investigate the trade-off between computational inference speed and the effectiveness of generating molecular structures Huang et al. [2023]. Because diffusion models inherently have high latency during inference, we attempt to explicitly quantify this trade-off curve.

Step-Budget Variable (T) We ablate the main experimental variable, the backsampling step size budget, and place it within a strictly defined range: $T \in \{25, 50, 100, 200\}$. By modifying only the **num_scales** parameter to correspond to T, we observe two extreme cases:

- $T = 25, 50$: The number of steps is greatly compressed, resulting in fast but potentially coarse and error-prone probability sampling.
- $T = 100, 200$: A slower integration path with higher granularity, minimizing truncation errors in the inverse stochastic differential equation (SDE) trajectory.

Sampling Protocol For every configuration of T, we apply an identical and strictly controlled generative protocol. We invoke the frozen checkpoint_200 model, iterating continuously to accurately sample 5,000 different molecular graphs. To optimize computational memory and prevent single-round timeouts during evaluation, all runs are enforced with an inference batch size of 16. After generation, the topology graph structure is converted into a one-dimensional SMILES string for subsequent robust cheminformatics evaluation and filtering.

3.5 Constraint Settings

Generative neural architectures output statistical probabilities mapping to discrete states; they do not inherently possess a physical understanding of chemical laws. Consequently, unguided discrete models frequently generate structurally impossible graphs. We address this by running our evaluation under two distinct analytical regimes.

1. **None (unconstrained baseline):** The raw output generated directly by the pre-trained baseline is evaluated without any post-hoc correction or filtration. This is the primary experimental mechanism for separating and analyzing the direct impact of the diffusion step size variance (T) on the raw generation capability.
2. **Filterplus (deterministic sanitization):** This constitutes our secondary appended experiment. Instead of changing the generator or retraining the model, we apply a sequential post-processing pipeline to a pool of 5000 generated SMILES strings. Each generated molecule must pass through the following 6 filtering steps in sequence. And figure 1 below illustrates the complete workflow for cleaning data with these six specific steps in a serial filtering process:
    - **RDKit Sanitization:** The molecule must pass fundamental computational chemistry validity checks via the RDKit open-source suite Bento et al. [2020].
    - **Atom Set Check:** The parsed graph must consist strictly of atoms $X_i \in \{C, N, O, F\}$.
    - **Connectivity Constraint:** The structure must represent a fully connected component. Discontinuous or non-intersecting graphs will be discarded.
    - **Size Constraint:** The total number of heavy atoms must be ≤ 9.
    - **Simple Valency Rules:** Molecules are checked according to the maximum valence state constraints: $C \leq 4$, $N \leq 4$, $O \leq 2$, and $F \leq 1$. Any molecule exceeding these local electronic bonding constraints is rejected.
    - **Canonicalization:** All molecules that successfully pass the above five checks will be normalized again using RDKit to achieve a unified evaluation.

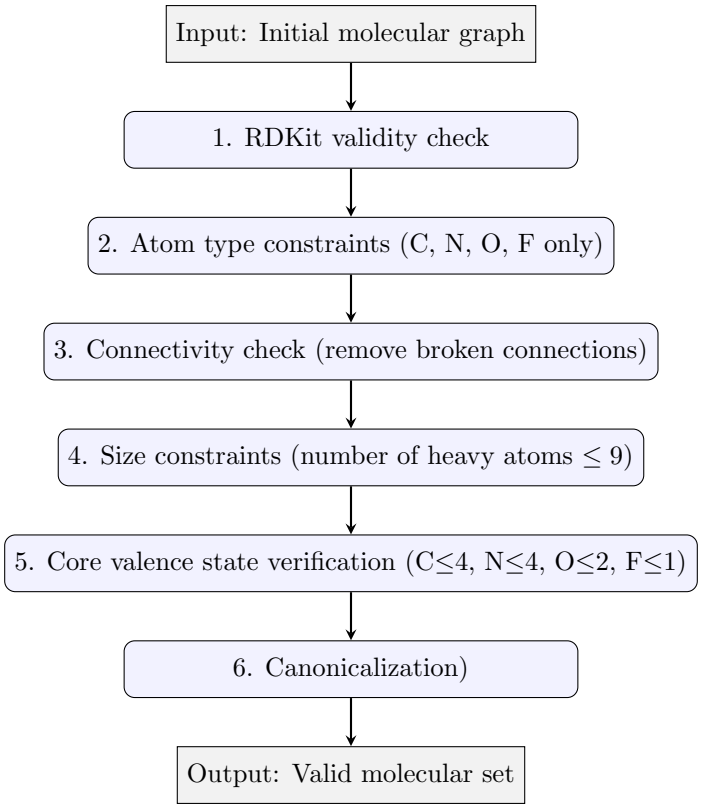

Figure 1: Filterplus serial post-processing pipeline

3.6 Evaluation Metrics

To robustly evaluate the model's performance under different T-step budget and constraint settings, we define core evaluation metrics. Let $\mathcal{G}_{input}$ represent the initial 5000 generated molecules, and $\mathcal{G}_{retained}$ represent the subset of molecules that successfully pass the above screening criteria.

- **Retention Ratio:** This metric calculates the proportion of generated molecule batches that are converted into physically stable and rule-compliant chemical structures. It is an indicator of absolute effectiveness:

$$\text{Retention Ratio} = \frac{|\mathcal{G}_{retained}|}{|\mathcal{G}_{input}|}$$

- **Unique Ratio:** This metric assesses the generative diversity of the model and identifies mode collapse. It is calculated by dividing the number of structurally unique molecules within the retained set by the total number of retained molecules.

$$\text{Unique Ratio} = \frac{|\text{Unique}(\mathcal{G}_{retained})|}{|\mathcal{G}_{retained}|}$$

- **FCD (Fréchet ChemNet Distance):** Evaluates the statistical similarity between the distribution of the generated molecules and the real-world QM9 test set by comparing their intermediate activations within a pre-trained ChemNet neural network.

- **Novelty:** Ensures the model's generative ability to synthesize entirely new compounds, rather than simply memorizing training data. It calculates the proportion of unique molecules retained that did not appear on the original training set.

- **Failure Breakdown (`valence_fail`):** In addition to successful generation, we track the specific reasons why molecules are rejected in the filtering process to understand how an aggressive step budget affects local structural constraints.
- **Runtime:** Since the architecture weights and batch size remain constant, changes in total running time directly reflect the additional sampling cost of increasing the total diffusion steps (T). This makes our analysis based on a practical trade-off between sampling efficiency and generation quality.

3.7 Project Teamwork

3.7.1 Resource Allocation

To complete this project, we first allocated workloads based on our respective strengths. Shiya Feng was responsible for the theoretical background and methodology. Binheng Zheng was primarily responsible for designing and conducting the experiments. Yuefeiyang Li was responsible for the environment setup and supporting the experimental pipeline. The three also provided cross-assistance to each other. We used GitHub for version control of the evaluation scripts and communicated offline almost daily. Due to the high memory requirements of the discrete diffusion model, our main physical resource constraint was access to cloud GPUs. We allocated GPU time sequentially, ensuring that evaluation runs with larger step budgets ($T = 100$ and $T = 200$) were scheduled for nighttime to maximize the use of our limited computing resources.

3.7.2 Actual vs. Estimated Workload

While our initial plan was that sample generation would be the most time-consuming phase, the actual workload was primarily concentrated in the post-processing and index evaluation stages. The debugging workload required to write the deterministic chemisensing filter pipeline far exceeded expectations. Teamwork proved to be far more efficient than working alone. This collaborative debugging approach saved us from several days of personal frustration.

3.7.3 Challenges and Solutions

Our main technical challenge was computational memory timeouts. During the initial sampling phase, when sampling 5000 molecules, the computation of the dense adjacency matrix during the reverse process repeatedly caused out-of-memory (OOM) crashes on our single-GPU configuration. We resolved this by dynamically adjusting our experimental protocol to enforce a strict inference batch size of 16. This increased the overall runtime but ensured stability. Furthermore, to ensure fair evaluation of our baseline CDGS checkpoints under different sampling budgets, we strictly managed hyperparameters. We avoided potential inconsistencies by establishing a unified, shared configuration file that all team members had to use before running their assigned evaluation batches.

4 Experimental Setup

[Primary author: Yuefeiyang Li]

To evaluate whether discrete molecular diffusion can achieve good generation quality, chemical efficiency, and structural novelty, we establish a rigorously controlled experimental environment. This chapter details the properties of the dataset, the preprocessing required to map chemical structures to graph theory space, computational hardware limitations, and specific hyperparameter configurations used for ablation studies.

4.1 Dataset Description

Our experiments use the QM9 dataset, a standard benchmark dataset for small molecule generation and property prediction in geometric deep learning Ramakrishnan et al. [2014]. This dataset comprises structures and properties of more than 1.3×10^5 stable and synthetically feasible small organic molecules. Geometries and structures are calculated using density functional theory (DFT) at the level of B3LYP/6-31G(2df,p). The initial geometries of these molecules were generated from SMILES strings reported in the GDB17 chemical universe Ruddigkeit et al. [2012].

In addition, the dataset is specifically designed for small drug-like molecules. Each molecule in the dataset contains a maximum of nine heavy atoms, specifically limited to the heavy-atom elements carbon (C), nitrogen (N), oxygen (O), and fluorine (F). We chose QM9 as the subject of this project because of its limited chemical space and relatively small graph size, which make it computationally tractable, allowing for ablation studies without the need for large industrial computing clusters. Out of these molecules in the QM9 set, 3054 have been labeled as uncharacterized because they rearranged during geometry optimization. This 3054 set has been curated Senthil et al. [2021] using a high-throughput workflow that performs CONNectivity preserving Geometry Optimization (ConnGO) with the same DFT method as in Ramakrishnan et al. [2014]. With ConnGO, 2830 molecules out of 3054 converge successfully to geometries with correct connectivities encoded in their original SMILES. Thus, the resulting QM9 dataset used in our experiments contains approximately 1.3×10^5 molecules.

In the context of discrete graph diffusion, QM9 provides the true distribution $P_{data}(G)$. The goal of the model is to learn this distribution so that the graph obtained from samples drawn from the trained model is almost indistinguishable from the real QM9 molecules. To ensure a fair comparison across sampling settings, we follow the standard preprocessing and dataset split used in the baseline setup. In our project, however, we do not retrain the model. Instead, we use a fixed pretrained checkpoint for evaluation under different sampling budgets and post-processing settings.

4.2 Data Preprocessing

Raw molecular data are stored as SMILES strings or 3D coordinate files (XYZ format). However, discrete diffusion models require data formatted as mathematical graphs. We employ a preprocessing pipeline consistent with the baseline setup to transform the raw QM9 data into graph representations with node and edge features.

Graph Construction For each molecule, we define a graph $G = (V, E)$, where V represents the set of nodes (atoms) and E represents the set of edges (bonds). Node features (X): Each atom is converted into a discrete category label. Following the baseline preprocessing used in our experiments, we use one-hot encoding over the atom types C, N, O, and F. Edge features (A): Bonds between atoms are represented using adjacency tensors. We know that in chemistry, bonds

are not simply connected or broken, but have discrete types. We encode edges into different categories: no bond (0), single bond (1), double bond (2), and triple bond (3).

RDKit-based Checks. During evaluation, we apply additional chemical validation where required. Under the none setting, the raw generated outputs are evaluated directly without additional post-processing. Under the filterplus setting, each generated SMILES is checked sequentially by RDKit sanitization, atom-type filtering (allowing only C, N, O, and F), single-fragment connectivity screening, simple valence-rule checks, and the maximum atom-count constraint. Molecules that fail any of these checks are discarded, while retained molecules are canonicalized before further analysis.

For the simple valence-rule check, we use the following upper bounds: carbon ≤ 4, nitrogen ≤ 4, oxygen ≤ 2, and fluorine ≤ 1.

4.3 Computational Environment

Although discrete diffusion models are mathematically efficient, they can be memory-intensive due to the need to compute transition probabilities on dense adjacency matrices during the reverse process. However, by limiting our experiments to QM9 and focusing on a fixed pretrained CDGS baseline and controlled evaluation experiments, we keep the computational demands manageable.

Hardware Specifications. Our experiments were conducted using cloud-based GPU resources available to the team, depending on availability and job duration. Since we focus on a fixed pretrained CDGS baseline on QM9 and controlled ablation experiments rather than large-scale training, a single-GPU setup is sufficient for running sampling and evaluation efficiently.

Software Stack. The core diffusion process and neural network architecture are implemented using PyTorch, together with PyTorch Geometric for efficient graph tensor operations. We use RDKit for cheminformatics processing, chemical validation, and metric-related evaluation.

Given the inherent randomness of diffusion-based generation, reproducibility remains an important concern. To mitigate this issue, we record software dependencies, configurations, and environment settings, and keep the evaluation setup consistent across different sampling budgets.

4.4 Hyperparameter Configuration

Our experimental design relies on controlled ablation to answer our main scientific questions. The most critical variable in this study is the sampling budget, defined as the number of diffusion steps T. All experiments are conducted under the CDGS baseline with VPSDE, using a fixed pretrained checkpoint, `checkpoint_200`. We do not retrain the model for different settings. Instead, we keep the generator fixed and vary only the sampling step budget and the post-processing condition.To make the setup easier to read, Table 3 summarizes the shared experimental settings used throughout all QM9 ablation experiments.

Step Budget Scan. In traditional diffusion methods, generating a single sample may require a large number of steps, which can become computationally expensive for high-throughput molecular generation. Our goal is to quantify the speed–quality trade-off by forcing the model to generate molecules in fewer steps. We define a step-budget scan, where $T \in \{25, 50, 100, 200\}$. A lower T implies a faster generation rate, but also a higher risk of degraded distributional quality and poorer retained-set reliability after filtering. In implementation, all other settings are kept fixed, and only `num_scales` is changed to match the target step budget T.

For each value of T, we sample from the same pretrained checkpoint and generate 5000 molecules. The evaluation batch size is fixed to 16 in order to keep each sampling round

manageable in runtime. The generated outputs are saved as SMILES strings for subsequent analysis.

Constraint Settings. In addition to the step budget, we also implement constraints as ex post-hoc variables. We consider two settings:

Table 2: Constraint settings used in the evaluation.

Setting	Description	Note
none	Directly evaluate the raw outputs generated by the baseline without additional post-processing.	Raw-output evaluation
filterplus	Apply a deterministic post-processing pipeline to the generated molecules without modifying the generator or retraining the model.	Lightweight chemistry-aware filtering

Under the filterplus setting, each generated SMILES is checked sequentially by RDKit sanitization, atom-type filtering (allowing only C, N, O, and F), single-fragment connectivity screening, simple valence-rule checks, and the maximum atom-count constraint. For the simple valence-rule check, we use the following upper bounds: carbon ≤ 4, nitrogen ≤ 4, oxygen ≤ 2, and fluorine ≤ 1. Molecules that fail any of these checks are discarded, while retained molecules are canonicalized before further analysis. This design allows us to study how lightweight chemistry-aware filtering interacts with the sampling budget in terms of reliability, diversity, and retention.

Table 3: Summary of the experimental setup used in all QM9 ablation experiments.

Parameter	Value	Note
Dataset	QM9	Small-molecule benchmark with heavy atoms in {C, N, O, F}
Baseline model	CDGS (VPSDE)	Fixed pretrained generator
Checkpoint	200	Same checkpoint for all evaluations
Sampling step budgets T	{25, 50, 100, 200}	Main ablation variable
Constraint settings	none, filterplus	Raw evaluation vs lightweight post-processing
Generated molecules per setting	5000	Same sample count for all settings
Evaluation batch size	16	Fixed across all runs
Filterplus rules	C, N, O, F; single fragment; heavy atoms ≤ 9; C≤ 4, N≤ 4, O≤ 2, F≤ 1	Applied only in filterplus

5 Experiments and Results

[Primary author: Binheng Zheng]

This section presents the experimental design and the corresponding empirical results. Our study is centered on a controlled step-budget sweep, in which the pretrained CDGS generator is kept fixed and only the number of diffusion sampling steps is varied. This design allows us to examine, in a systematic way, how the sampling budget influences generation quality, diversity, and computational efficiency.

5.1 Main Experiment: Step-Budget Sweep

The primary experiment in this project is a step-budget sweep over the number of diffusion sampling steps. Throughout this experiment, we keep the pretrained CDGS model fixed and vary only the sampling budget in order to compare generation behavior under different inference settings. Specifically, we consider four step budgets:

$$T \in \{25, 50, 100, 200\}.$$

All settings use the same pretrained model checkpoint, `checkpoint_200`, so that differences across experiments can be attributed as directly as possible to the sampling procedure itself. For each value of T, we generate 5000 molecules under the same sampling protocol, with the evaluation batch size fixed to 16. The generated outputs are saved as SMILES strings for subsequent evaluation and analysis.

This experimental design is motivated by the role of the sampling budget in diffusion-based generation. In general, a larger value of T corresponds to a more gradual denoising trajectory, which is expected to produce more stable and higher-quality samples, but at the cost of increased inference time. Conversely, a smaller value of T yields a more compressed and computationally cheaper sampling process, though potentially with a loss in generation quality. The step-budget sweep therefore provides a direct way to examine how changes in T affect both efficiency and output quality.

This choice is also well aligned with the first research question of the project, namely how the sampling budget influences runtime and generation performance. Because the generator, checkpoint, and evaluation pipeline are held fixed across all settings, the resulting comparisons are easier to interpret than comparisons that would additionally involve changes in training, architecture, or optimization.

In the main experiment, we use the **none** setting as the primary evaluation setting. Under this configuration, the raw outputs produced by the baseline model are evaluated directly, without any additional post-processing. This makes it possible to isolate the effect of step-budget variation itself before introducing any chemistry-aware filtering.

5.2 Auxiliary Experiment: Filterplus Post-Processing

In addition to the step-budget sweep, we conduct an auxiliary experiment to examine the effect of lightweight chemistry-aware post-processing. Whereas the main experiment is designed to reveal the raw behavior of the fixed generator under different sampling budgets, this auxiliary experiment shifts the focus to the retained molecular set after post-generation screening. The aim is not to modify the model, alter the training procedure, or introduce a sophisticated repair mechanism, but rather to investigate whether a simple rule-based filtering stage can improve the reliability of the generated outputs. Concretely, we compare two constraint settings:

$$\text{constraint} \in \{\text{none}, \text{filterplus}\}.$$

Under the none setting, the raw model outputs are evaluated directly, without any additional post-processing. Under the filterplus setting, each generated SMILES string is subjected to a sequence of deterministic chemical checks before evaluation. In the final implementation, the filterplus pipeline applies the following steps in order:

1. RDKit sanitization,
2. atom-type filtering, allowing only C, N, O, and F,
3. single-fragment connectivity check,
4. simple valence-rule check,
5. maximum atom-count constraint,
6. canonicalization of retained molecules.

For the valence-rule check, we use simple upper bounds consistent with the QM9 setting: carbon ≤ 4, nitrogen ≤ 4, oxygen ≤ 2, and fluorine ≤ 1. Molecules that fail any one of these checks are discarded, while the retained molecules are canonicalized and passed to downstream analysis.

The central question of this auxiliary experiment is whether such lightweight chemistry-aware filtering can improve reliability without incurring an excessive loss of diversity or sample retention. In this sense, filterplus should be understood as a conservative post-processing strategy rather than a repair algorithm. It does not attempt to correct invalid molecules or modify the sampling trajectory itself; instead, it removes outputs that violate basic chemical constraints. This makes the procedure easy to implement, computationally inexpensive, and methodologically consistent with the scope of the project.

This design also serves a clear analytical purpose. By comparing none and filterplus under the same value of T, we can distinguish improvements that arise from post-processing from those that arise from the sampling budget itself. Across all experiments, the full design therefore consists of four step settings and two constraint settings, yielding eight core experimental conditions. Comparisons across T reveal how the sampling budget affects speed and generation quality, whereas comparisons between none and filterplus at fixed T show whether lightweight post-processing improves reliability and what retention cost is paid in return.

5.3 Evaluation Metrics

To compare different experimental settings in a consistent and interpretable way, we report a set of metrics that capture generation quality, diversity, efficiency, and post-processing behavior. Because the task is molecular graph generation rather than generic graph generation, the most informative metrics are those that reflect both chemical plausibility at the sample level and usefulness of the final retained set at the distribution level.

We begin with **Validity**, which measures whether a generated molecule is chemically valid after conversion into a molecular object and application of the required validation procedure. Under the none setting, validity reflects the raw generation quality of the baseline model. Under the filterplus setting, it reflects the validity of the retained set after deterministic chemistry-aware screening. Since one of the main questions of this project is whether lightweight filtering can improve chemical reliability, validity remains a central quality indicator throughout the study.

As a complementary distribution-level metric, we report **FCD/Test**. This metric compares the generated molecules with the reference molecules in the test set and measures how closely the generated distribution matches the target molecular distribution. Whereas validity focuses on whether individual samples satisfy chemical constraints, FCD/Test evaluates the overall realism of the generated set from a distributional perspective. The two metrics therefore capture different aspects of quality: one is local and sample-specific, while the other is global and distributional.

We also report **Novelty**, defined as the proportion of retained molecules that do not appear in the training set. Novelty is useful as an indicator of whether the generator is merely reproducing known molecules or is able to produce unseen outputs. At the same time, its interpretation on QM9 requires caution. As noted in the CDGS paper, QM9 is close to a near-enumeration of small molecules under predefined constraints, so a higher novelty score does not automatically correspond to a better generative model. In our analysis, novelty is therefore treated as a supporting diversity metric rather than as a standalone measure of model quality.

To characterize repetition more directly, we place particular emphasis on retained-set statistics arising from the filterplus pipeline. Specifically, we report **Unique filtered count**, which denotes the number of unique molecules remaining after filtering and canonicalization, and **Unique ratio**, defined as

$$\text{Unique ratio} = \frac{\text{number of unique retained molecules}}{\text{number of retained molecules}}.$$

This ratio quantifies the degree of repetition within the retained set. It is especially informative under post-processing, because an apparent gain in validity is less meaningful if the surviving molecules are highly repetitive.

A closely related quantity is the **Retention ratio**, defined as

$$\text{Retention ratio} = \frac{\text{number of retained molecules}}{\text{number of input molecules}}.$$

This metric is crucial for interpreting the real effect of filterplus post-processing. In principle, a filtering strategy can improve the quality of the final set simply by discarding a large number of problematic outputs. The retention ratio therefore provides the necessary context for judging whether an improvement in validity or FCD/Test is achieved at an acceptable cost in sample survival.

In addition, we report a **failure breakdown** for the filterplus pipeline, with particular attention to categories such as `valence_fail`. Because the post-processing checks are applied sequentially, this breakdown reveals which chemical constraints are most frequently violated by the raw outputs. It is therefore useful not only for understanding the behavior of the filter itself, but also for diagnosing how the weaknesses of the generator vary across different sampling budgets.

Finally, we report **Runtime** as the primary efficiency metric. Since one of the central questions of the project is how much computational cost can be reduced by compressing the sampling budget, runtime provides the most direct measure of sampling efficiency. In our setup, all factors other than the step budget T are fixed, so differences in runtime can be interpreted directly as the additional sampling cost associated with using more diffusion steps.

5.4 Results under the None Setting

5.4.1 Overall Quantitative Results

We begin with the main results under the none setting, in which the raw outputs of the fixed CDGS baseline are evaluated directly without any additional post-processing. This configuration is intended to isolate the effect of the sampling step budget T while keeping all other factors as stable as possible. Since all runs use the same pretrained checkpoint, sampling protocol, number of generated molecules, and evaluation batch size, the observed differences can be attributed primarily to the sampling budget itself rather than to changes in training, architecture, or evaluation procedure.

In the main experiment, we compare four sampling-step settings, namely $T \in \{25, 50, 100, 200\}$. These settings span a range from faster but coarser sampling to slower but more refined sampling, and are therefore well suited to examining how the quality of discrete diffusion molecular

generation changes as the inference budget increases. For each value of T, we generate 5000 molecules from the same fixed checkpoint and evaluate the outputs directly. Because the none setting contains no additional filtering stage, it provides the clearest view of the generator's raw behavior under different sampling budgets.

Table 4: Main quantitative results under the none setting for four sampling step budgets. All 5000 generated molecules are evaluated directly without additional post-processing.

T	#Samples	Validity	Unique ratio	Novelty	FCD/Test
25	5000	1.0	0.7654	0.994251	9.843182
50	5000	1.0	0.7582	0.993142	9.434804
100	5000	1.0	0.8578	0.953835	6.028562
200	5000	1.0	0.9868	0.723551	0.272468

Table 4 summarizes the overall quantitative results under the none setting. Several patterns are immediately visible. First, *Validity* remains at 1.0 across all four step budgets, indicating that, under the evaluation pipeline used here, all retained outputs satisfy the validity criterion. Second, *FCD/Test* decreases dramatically as T increases, dropping from 9.84 at $T = 25$ to 0.27 at $T = 200$. This large improvement indicates that larger step budgets produce samples whose overall distribution is substantially closer to that of the QM9 test set. Third, the *Unique ratio* rises from roughly 0.76 at the two smallest budgets to nearly 0.99 at $T = 200$, which suggests that increasing the number of sampling steps markedly reduces repetition within the generated set. By contrast, *Novelty* declines as T increases. In the present context, however, this decrease should be interpreted with caution, since novelty on QM9 is not a standalone indicator of model quality and must be considered jointly with distributional quality and repetition.

5.4.2 Efficiency–Quality Trade-off

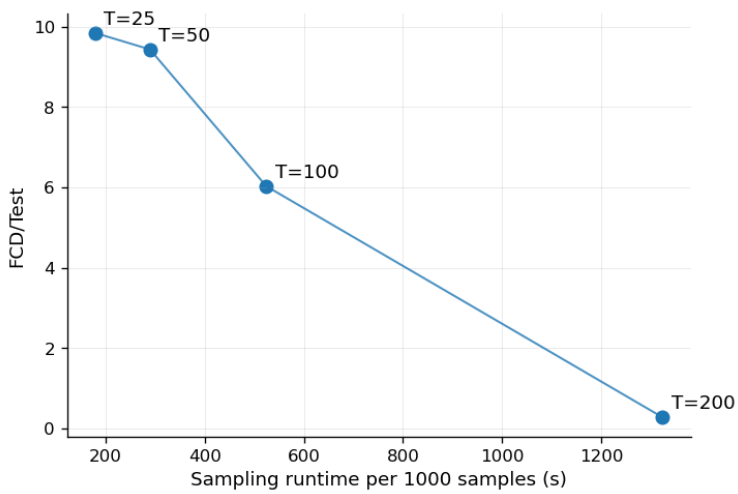

Figure 2: Pareto-style scatter of sampling runtime per 1000 samples versus FCD/Test under the none setting. Each point corresponds to one sampling step budget T. Larger step budgets improve distributional quality but increase sampling cost.

Figure 2 makes the relationship between efficiency and distributional quality particularly explicit by plotting sampling runtime per 1000 samples against FCD/Test. The pattern is clear and consistent with the numerical results in Table 4. The points associated with $T = 25$ and $T = 50$ lie in the low-runtime region, confirming their computational advantage, but they are also associated with substantially larger FCD/Test values, which indicates weaker agreement with the target molecular distribution. As T increases to 100 and then 200, the points move steadily

toward lower FCD/Test, but at the cost of substantially increased runtime. In particular, $T = 200$ occupies the most expensive part of the plot while also achieving the best distribution-level quality by a wide margin.

This figure therefore visualizes the central trade-off of the none-setting experiment: aggressive compression of the sampling budget yields clear efficiency gains, but these gains are accompanied by a pronounced loss in generation quality. The relationship is not marginal. The difference between $T = 25$ and $T = 200$ is large both in runtime and in FCD/Test, which suggests that the choice of sampling budget has a first-order effect on the practical behavior of the generator.

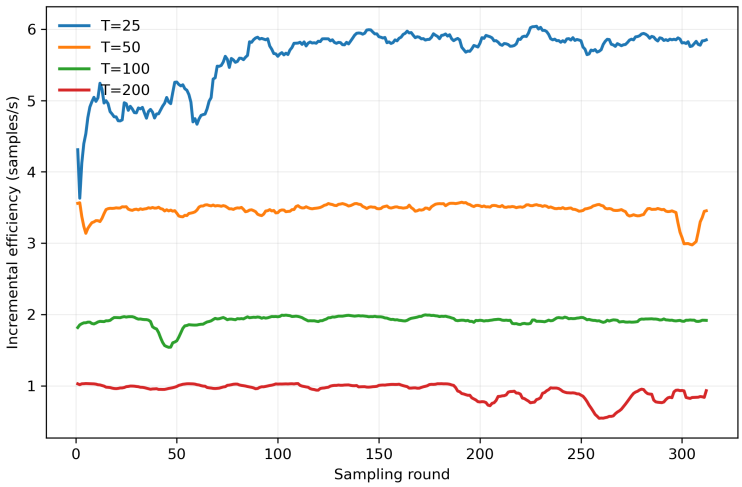

Figure 3: Incremental sampling efficiency over the generation process under the none setting, measured in samples per second at each sampling round. Larger step budgets show consistently lower efficiency throughout the process, which helps explain the runtime differences observed in Figure 2.

Figure 3 complements the runtime–quality scatter by providing a process-level view of sampling efficiency. Instead of reporting only the final runtime, it traces incremental efficiency during generation, measured as the number of generated samples per second at each sampling round. This representation helps explain *how* the runtime gap emerges. The figure shows that larger step budgets remain less efficient across nearly the entire generation process, whereas smaller values of T maintain consistently higher throughput. In other words, the higher total runtime of large-T sampling is not caused by a few isolated slow rounds, but by a systematic reduction in per-round efficiency throughout the procedure.

Taken together, Figures 2 and 3 provide two complementary views of the same phenomenon. The former summarizes the final efficiency–quality trade-off, while the latter reveals the mechanism by which the efficiency gap accumulates over time. This strengthens the conclusion that larger sampling budgets improve quality precisely because they allocate substantially more computational effort across the entire denoising process.

5.4.3 Diversity Analysis

Figure 4 focuses on diversity-related behavior by plotting Novelty and Unique ratio as functions of the sampling step budget. The two curves reveal a clear divergence. On the one hand, the steady increase in Unique ratio indicates that larger step budgets produce less repetitive sets of generated molecules. This is consistent with Table 4, where the unique ratio rises sharply from the low-T settings to nearly one at $T = 200$. On the other hand, Novelty decreases as T grows, meaning that the higher-quality and less repetitive outputs at larger step budgets also lie closer to molecules already represented in the training distribution.

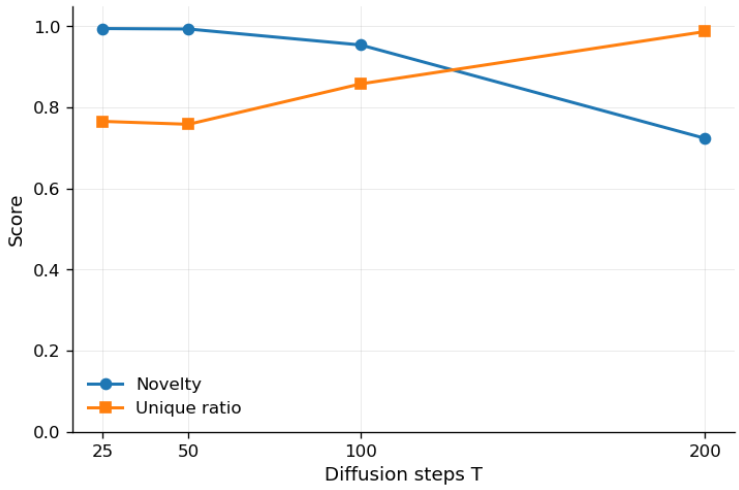

Figure 4: Novelty and Unique ratio as functions of the sampling step budget T under the none setting. Larger step budgets yield less repetitive outputs, while novelty decreases across the tested range.

This contrast is informative. Increasing T does not improve all diversity-related quantities simultaneously. Rather, it improves one aspect of diversity—namely non-redundancy within the generated set—while reducing another, namely the fraction of molecules not seen during training. In the QM9 setting, this should not be read as a contradiction. Because novelty is known to be difficult to interpret on a nearly enumerated small-molecule dataset, the more stable conclusion is that larger step budgets reduce repetition and improve distributional quality, while novelty serves mainly as a supporting statistic rather than a decisive criterion.

Overall, the none-setting results present a coherent picture. Larger step budgets substantially improve FCD/Test and reduce repetition, but they do so at a clear computational cost. Smaller budgets provide faster generation, yet this speed advantage is accompanied by noticeably worse distributional quality and stronger redundancy in the generated set. The none-setting experiment therefore establishes the baseline efficiency–quality relationship against which the subsequent filterplus results can be interpreted.

5.5 Results under the Filterplus Setting

5.5.1 Retained-Set Statistics

We now turn to the results under the filterplus setting. Unlike the none configuration, this setting does not evaluate the raw generated outputs directly. Instead, each generated SMILES string is passed through the lightweight chemistry-aware post-processing pipeline described above, and only the molecules that survive all checks are retained for subsequent analysis. The purpose of this experiment is therefore not to improve the generator itself, but to assess whether a simple deterministic filtering stage can make the final molecular set more reliable, and at what cost in terms of sample retention.

Because the same post-processing rules are applied across all step budgets, the filterplus results provide a useful view of reliability after screening. In particular, they allow us to distinguish between two different questions: whether a given step budget produces high-quality raw outputs, and whether those outputs remain useful after basic chemical constraints are enforced. This shift in perspective is important, since the final quality of a retained set depends not only on how well the generator performs, but also on how many samples are lost during filtering.

Table 5 summarizes the retained-set statistics under the filterplus setting. Several patterns are immediately visible. Most importantly, the *Retention ratio* increases strongly with the sampling

Table 5: Post-processing results under the filterplus setting for four sampling step budgets. The table reports retained-set statistics and valence-related failures after lightweight chemistry-aware filtering.

T	Filtered count	Dropped count	Retention ratio	Unique ratio	Novelty	FCD/Test	Valence fail
25	2753	2247	0.5506	0.745369	0.989279	8.974123	2247
50	3360	1640	0.6720	0.733631	0.989452	8.832106	1640
100	4158	842	0.8316	0.844637	0.943622	5.736158	842
200	4935	65	0.9870	0.986626	0.719860	0.264977	65

budget, from 0.5506 at $T = 25$ to 0.9870 at $T = 200$. This shows that larger step budgets produce substantially fewer molecules that fail the post-processing pipeline. The same pattern appears in the absolute counts: the number of dropped molecules falls sharply from 2247 at $T = 25$ to only 65 at $T = 200$. In other words, as the denoising process becomes more complete, the raw outputs are increasingly likely to satisfy even these simple chemical constraints.

The retained-set quality also improves with T. The *FCD/Test* value decreases steadily across the four settings, reaching 0.264977 at $T = 200$, which is very close to the corresponding none-setting result. At the same time, the *Unique ratio* rises from approximately 0.74 at the smallest budgets to 0.986626 at $T = 200$, indicating that the retained molecules become substantially less repetitive as the sampling budget increases. Novelty follows the same downward trend already observed under the none setting and should again be interpreted with caution on QM9. Taken together, the table suggests that larger step budgets not only improve the raw outputs, but also make those outputs much more likely to survive post-processing with relatively little loss.

5.5.2 Comparison with the None Setting

From the perspective of the second research question, Table 5 is especially informative because it makes clear that post-processing gains must always be interpreted together with retention loss. An apparent improvement in the final retained set is less convincing if it is achieved only by discarding a large fraction of the generated molecules. This trade-off is particularly visible at small step budgets: although filtering improves the quality of the surviving set, the retention loss is substantial at $T = 25$ and $T = 50$. By contrast, at $T = 200$, the retained-set quality remains strong while almost the entire sample pool is preserved.

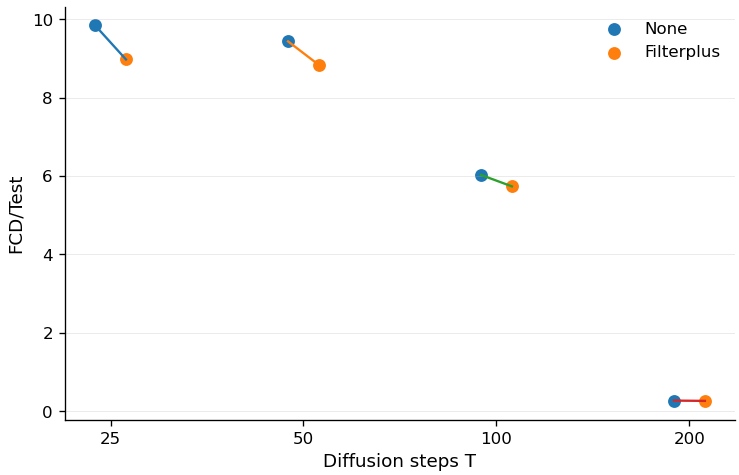

Figure 5: Comparison of FCD/Test under the none and filterplus settings across four sampling step budgets. Filterplus consistently yields slightly lower FCD/Test values, but does not change the overall ranking across T.

Figure 5 compares the distributional quality of the none and filterplus settings through FCD/Test. For every value of T, the filterplus curve lies slightly below the none curve, which indicates that lightweight chemistry-aware filtering consistently improves the quality of the final retained molecular set. The effect is systematic rather than accidental: the improvement appears at all four step budgets. At the same time, the size of the improvement is modest, and the global ordering across T remains unchanged. In particular, $T = 200$ still performs best, followed by $T = 100$, while $T = 25$ and $T = 50$ remain the weakest settings.

This result is important because it clarifies what filterplus can and cannot do. It can improve the quality of the retained set by removing molecules that violate basic chemical constraints, but it does not alter the underlying dependence of generation quality on the sampling budget itself. In other words, post-processing refines the outputs, but it does not compensate for an insufficient sampling budget. The quality hierarchy across different values of T is therefore fundamentally determined by the generator and the sampling process, not by the filtering stage.

5.5.3 Failure Breakdown

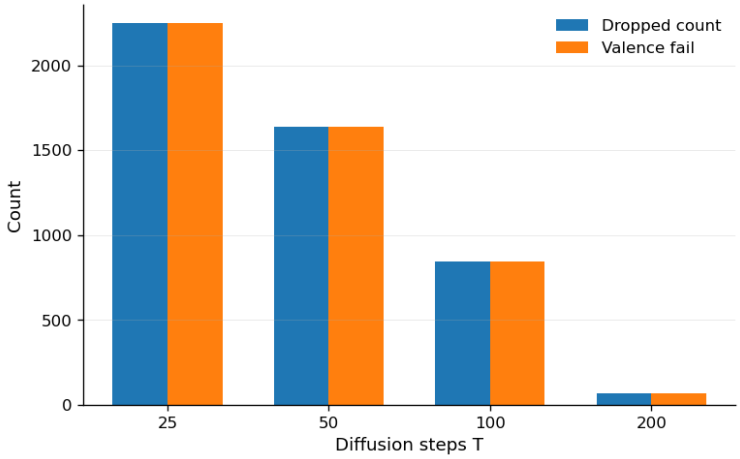

Figure 6: Dropped-count and valence-fail statistics across four sampling step budgets under the filterplus setting. The near-complete overlap of the two bars indicates that valence-rule violations dominate sample removal in the final implementation.

Figure 6 provides a more detailed view of why molecules are removed under filterplus. The most striking feature is the near-perfect overlap between the dropped-count bars and the valence-fail bars for all four step budgets. This indicates that, in the final implementation, almost all discarded molecules are removed because of valence-rule violations rather than atom-type, connectivity, or atom-count failures. The filtering pipeline therefore acts mainly as a valence-consistency screen in practice.

The figure also shows that both dropped count and valence-fail count decrease sharply as T increases. This trend is highly informative. It suggests that larger step budgets do not merely improve final FCD/Test; they also reduce the number of chemically problematic outputs at the source. In other words, the benefit of increasing T is visible not only in distributional quality, but also in the chemical reliability of the raw samples before post-processing. This observation strengthens the interpretation of the previous table and figure: the high retention ratio at large T is not accidental, but reflects a genuine improvement in the generator's ability to produce chemically consistent molecules when allowed a more complete denoising process.

Taken together, Table 5 and Figures 5–6 provide a coherent picture of the filterplus results. Table 5 shows how much of the generated sample pool survives filtering and how the retained-set quality changes with T. Figure 5 demonstrates that post-processing yields a consistent but

moderate improvement in FCD/Test, while Figure 6 explains that this improvement comes mainly from eliminating molecules that violate simple valence constraints. Overall, filterplus emerges as a useful but conservative post-processing strategy: it improves the final retained set, but it does so primarily by discarding chemically problematic molecules rather than by changing the generator itself. This makes it a natural extension of the none-setting analysis rather than a separate modeling contribution.

5.6 Discussion and Key Findings

Several conclusions emerge from the experiments. First, the sampling budget plays a decisive role in shaping the behavior of the fixed CDGS baseline. Increasing T consistently improves distributional alignment with the QM9 test set and reduces redundancy in the generated molecules, but these gains come with a substantial increase in computational cost. The results therefore indicate that the choice of T is not a minor implementation detail, but a central factor governing the balance between efficiency and output quality.

Second, the filterplus experiment shows that lightweight post-processing is useful mainly as a screening mechanism. Its benefit lies in producing a cleaner retained set, not in compensating for weak raw generation. The improvements after filtering remain relatively modest, and they depend strongly on the quality of the initial samples. In particular, the failure analysis suggests that chemically inconsistent outputs are concentrated in a small number of simple violation types, with valence-related failures dominating in the final implementation.

More broadly, the experiments suggest that reliable molecular generation on QM9 is shaped by two complementary stages: the quality of the sampling trajectory itself, and the strictness of the criteria used to retain or discard outputs afterward. Within this framework, the main contribution of the project is a controlled empirical analysis of a fixed pretrained baseline, showing how sampling-budget adjustment and simple chemistry-aware filtering jointly affect the final results.

6 Conclusions and Outlook

[Primary author: Binheng Zheng]

This project presented a controlled empirical study of CDGS-based molecular generation on QM9, with a particular focus on the role of the diffusion sampling budget and the effect of lightweight chemistry-aware post-processing. By fixing the pretrained generator and varying only the number of sampling steps, we were able to examine the influence of inference-time decisions without introducing additional confounding factors from training or architecture changes.

The results show that the sampling budget is a major determinant of generation behavior. Larger step budgets consistently yield better distributional quality and lower redundancy in the generated set, but these gains come at a clear computational cost. Conversely, smaller step budgets improve efficiency, yet the reduction in runtime is accompanied by noticeably weaker alignment with the target distribution. This confirms that, even for a fixed pretrained baseline, the choice of sampling budget has a substantial effect on the final practical performance of the generator.

The filterplus experiment provides a complementary perspective. Lightweight post-processing does improve the quality of the retained set, but its effect remains limited and should be interpreted conservatively. In our implementation, the improvement comes primarily from removing chemically problematic outputs rather than repairing them, with valence-related violations accounting for most rejections. This suggests that simple filtering can be useful as a final screening stage, but it cannot substitute for strong raw generation quality.

Taken together, the experiments clarify the respective roles of sampling and post-processing in this setting. The sampling budget determines the overall quality–efficiency regime in which the generator operates, while lightweight chemistry-aware filtering offers a modest refinement of the final retained outputs. The main contribution of the project is therefore not methodological novelty in model design, but a focused empirical analysis of how these two factors jointly shape the behavior of a fixed pretrained molecular diffusion baseline on QM9.

A realistic direction for future work is to test whether the same patterns persist on datasets beyond QM9, where the interpretation of novelty and chemical validity may differ. It would also be worthwhile to explore slightly richer post-processing strategies that remain lightweight but can move beyond simple rejection-based filtering. Finally, further investigation of sampling acceleration methods may help determine whether comparable quality can be maintained under a reduced inference budget.

References

Jacob Austin, Daniel D Johnson, Jonathan Ho, Daniel Tarlow, and Rianne Van Den Berg. Structured denoising diffusion models in discrete state-spaces. *Advances in neural information processing systems*, 34:17981–17993, 2021.

A Patrícia Bento, Anne Hersey, Eloy Félix, Greg Landrum, Anna Gaulton, Francis Atkinson, Louisa J Bellis, Marleen De Veij, and Andrew R Leach. An open source chemical structure curation pipeline using rdkit. *Journal of cheminformatics*, 12(1):51, 2020.

Yoann Boget. Simple and critical iterative denoising: A recasting of discrete diffusion in graph generation. *arXiv preprint arXiv:2503.21592*, 2025.

Michael Cardei, Jacob K Christopher, Bhavya Kailkhura, Thomas Hartvigsen, and Ferdinando Fioretto. Constrained molecular generation with discrete diffusion for drug discovery. In *NeurIPS 2025 Workshop on AI Virtual Cells and Instruments: A New Era in Drug Discovery and Development*, 2025.

Hongyang Chen, Can Xu, Lingyu Zheng, Qiang Zhang, and Xuemin Lin. Diffusion-based graph generative methods. *IEEE Transactions on Knowledge and Data Engineering*, 36(12): 7954–7972, 2024.

Xiaohui Chen, Jiaxing He, Xu Han, and Li-Ping Liu. Efficient and degree-guided graph generation via discrete diffusion modeling. *arXiv preprint arXiv:2305.04111*, 2023.

Justin Gilmer, Samuel S Schoenholz, Patrick F Riley, Oriol Vinyals, and George E Dahl. Neural message passing for quantum chemistry. In *International conference on machine learning*, pages 1263–1272. Pmlr, 2017.

Aditya Grover, Aaron Zweig, and Stefano Ermon. Graphite: Iterative generative modeling of graphs. In *International conference on machine learning*, pages 2434–2444. PMLR, 2019.

Kilian Konstantin Haefeli, Karolis Martinkus, Nathanaël Perraudin, and Roger Wattenhofer. Diffusion models for graphs benefit from discrete state spaces. *arXiv preprint arXiv:2210.01549*, 2022.

Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *Advances in neural information processing systems*, 33:6840–6851, 2020.

Han Huang, Leilei Sun, Bowen Du, and Weifeng Lv. Conditional diffusion based on discrete graph structures for molecular graph generation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pages 4302–4311, 2023.

Jaehyeong Jo, Seul Lee, and Sung Ju Hwang. Score-based generative modeling of graphs via the system of stochastic differential equations. In *International conference on machine learning*, pages 10362–10383. PMLR, 2022.

Lingkai Kong, Jiaming Cui, Haotian Sun, Yuchen Zhuang, B Aditya Prakash, and Chao Zhang. Autoregressive diffusion model for graph generation. In *International conference on machine learning*, pages 17391–17408. PMLR, 2023.

Raghunathan Ramakrishnan, Pavlo O Dral, Matthias Rupp, and O Anatole Von Lilienfeld. Quantum chemistry structures and properties of 134 kilo molecules. *Scientific data*, 1(1):1–7, 2014.

Lars Ruddigkeit, Ruud Van Deursen, Lorenz C Blum, and Jean-Louis Reymond. Enumeration of 166 billion organic small molecules in the chemical universe database gdb-17. *Journal of chemical information and modeling*, 52(11):2864–2875, 2012.

Salini Senthil, Sabyasachi Chakraborty, and Raghunathan Ramakrishnan. Troubleshooting unstable molecules in chemical space. *Chemical science*, 12(15):5566–5573, 2021.

Yang Song, Jascha Sohl-Dickstein, Diederik P Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. *arXiv preprint arXiv:2011.13456*, 2020.

Clement Vignac, Igor Krawczuk, Antoine Siraudin, Bohan Wang, Volkan Cevher, and Pascal Frossard. Digress: Discrete denoising diffusion for graph generation. *arXiv preprint arXiv:2209.14734*, 2022.

Zhe Xu, Ruizhong Qiu, Yuzhong Chen, Huiyuan Chen, Xiran Fan, Menghai Pan, Zhichen Zeng, Mahashweta Das, and Hanghang Tong. Discrete-state continuous-time diffusion for graph generation. *Advances in Neural Information Processing Systems*, 37:79704–79740, 2024.

Chengxuan Ying, Tianle Cai, Shengjie Luo, Shuxin Zheng, Guolin Ke, Di He, Yanming Shen, and Tie-Yan Liu. Do transformers really perform badly for graph representation? *Advances in neural information processing systems*, 34:28877–28888, 2021.

Jiaxuan You, Rex Ying, Xiang Ren, William Hamilton, and Jure Leskovec. Graphrnn: Generating realistic graphs with deep auto-regressive models. In *International conference on machine learning*, pages 5708–5717. PMLR, 2018.

Xiaochen Zhang, Shuangxi Wang, Ying Fang, and Qiankun Zhang. Mg-diff: A novel molecular graph diffusion model for molecular generation and optimization. *Plos one*, 20(10):e0331450, 2025.